Vasicek Interest Rate Model: Theory, Calibration, and Backtesting

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1. Problem Statement

Interest rate modeling is fundamental in quantitative finance. It underpins the pricing of bonds, interest rate derivatives, mortgage-backed securities, and the management of fixed-income portfolios. A key challenge is that interest rates exhibit two empirically observed behaviors that distinguish them from stock prices:

1. **Mean reversion** — rates tend to drift back toward a long-run equilibrium level. When rates are very high (e.g., 20% in 1981), economic forces push them down. When rates are very low (e.g., near 0% in 2009–2015), they eventually rise again.

2. **Stochastic fluctuations** — despite mean reversion, rates are subject to continuous random shocks driven by monetary policy decisions, macroeconomic surprises, and market sentiment.

This report applies the **Vasicek (1977)** model — one of the first and most analytically tractable short-rate models — to the US Federal Funds Effective Rate. We:

- Derive the model and its analytical properties
- Estimate parameters using OLS and Maximum Likelihood (MLE)
- Validate the implementation on synthetic data with known ground truth
- Apply the model to 30 years of real Fed Funds Rate data and evaluate forecast quality

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2. Theory: The Vasicek Model

2.1 Stochastic Differential Equation

The Vasicek model describes the instantaneous short rate $r(t)$ as an Ornstein-Uhlenbeck (OU) process:

$$dr(t) = \kappa(\theta - r(t))\,dt + \sigma\,dW(t)$$

where:

Symbol	Name	Interpretation
$\kappa > 0$	Mean reversion speed	Rate at which $r(t)$ reverts to θ
θ	Long-run mean	The equilibrium rate that $r(t)$ gravitates toward
$\sigma > 0$	Volatility	Magnitude of random shocks per unit time
$W(t)$	Standard Brownian motion	Source of randomness

The drift term $\kappa(\theta - r(t))$ acts as a restoring force: - If $r(t) > \theta$: drift is negative $\rightarrow$ rate is pushed downward - If $r(t) < \theta$: drift is positive $\rightarrow$ rate is pushed upward - Strength of pull is proportional to the distance $|\theta - r(t)|$

2.2 Exact Solution

The SDE has a closed-form solution. Given $r(0) = r_0$:

$$r(t) = r_0\,e^{-\kappa t} + \theta(1 - e^{-\kappa t}) + \sigma \int_0^t e^{-\kappa(t-s)}\,dW(s)$$

The stochastic integral is Gaussian, so $r(t)$ is normally distributed with:

$$\mathbb{E}[r(t) \mid r_0] = r_0\,e^{-\kappa t} + \theta(1 - e^{-\kappa t})$$

$$\mathrm{Var}[r(t) \mid r_0] = \frac{\sigma^2}{2\kappa}\,(1 - e^{-2\kappa t})$$

Long-run behavior (as $t \rightarrow \infty$):

$$r(t) \xrightarrow{d} \mathcal{N}\left(\theta,\ \frac{\sigma^2}{2\kappa}\right)$$

The rate converges in distribution to a Gaussian with mean θ and variance $\sigma^2/(2\kappa)$.

2.3 Key Analytical Properties

Half-life of mean reversion:

$$t_{1/2} = \frac{\ln 2}{\kappa}$$

This is the time for a deviation from θ to shrink by half. A small κ implies slow mean reversion (rates can stay far from θ for years); a large κ implies fast reversion.

Discrete-time (Euler-Maruyama) approximation:

For estimation from discrete data with time step Δt, the exact conditional distribution is:

$$r(t + \Delta t) \mid r(t) \ \sim\ \mathcal{N}(\mu_t,\ v)$$

where:

$$\mu_t = r(t)\, e^{-\kappa \Delta t} + \theta(1 - e^{-\kappa \Delta t})$$

$$v = \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa \Delta t}\right)$$

2.4 Known Limitations

Limitation	Description
Negative rates	Since $r(t)$ is Gaussian, it can go negative with positive probability. Historically a flaw; less controversial after negative rate policies post-2008.
Constant parameters	κ, θ, σ are fixed. In reality, the Fed's long-run target shifts over time.
Single factor	Only one source of randomness. Cannot capture the full yield curve dynamics.

Limitation	Description
No regime shifts	Cannot model sudden, large policy changes (e.g., the 2022 rate hike cycle).

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3. Parameter Estimation

3.1 OLS Estimation

Rewriting the discretized SDE as a linear regression:

$$\underbrace{r(t+\Delta t) - r(t)}_{\Delta r} = \underbrace{\kappa\theta\,\Delta t}_{A} + \underbrace{(-\kappa\,\Delta t)}_{B}\; r(t) + \varepsilon_t$$

OLS regression of Δr on $r(t)$ gives coefficients $\hat{A}$ and $\hat{B}$, from which:

$$\hat{\kappa} = -\frac{\hat{B}}{\Delta t}, \qquad \hat{\theta} = -\frac{\hat{A}}{\hat{B}}, \qquad \hat{\sigma} = \frac{\mathrm{std}(\hat{\varepsilon})}{\sqrt{\Delta t}}$$

OLS is fast and closed-form but carries a known **upward bias** in $\hat{\kappa}$ for finite samples — it overestimates mean reversion speed, leading to confidence bands that are too narrow.

3.2 Maximum Likelihood Estimation (MLE)

MLE uses the exact conditional Gaussian distribution at each time step. The log-likelihood is:

$$\ell(\kappa, \theta, \sigma) = -\frac{N}{2}\ln(2\pi v) - \frac{1}{2v}\sum_{t=1}^{N}\left(r_t - \mu_{t-1}\right)^2$$

where μ_{t-1} and v use the exact expressions from Section 2.3. MLE is more accurate than OLS and is the recommended method. OLS estimates are used as the starting point for the numerical optimizer.

3.3 Finite-Sample Bias in κ

A well-documented property of mean reversion estimation is that both OLS and MLE **overestimate** κ in finite samples. For highly persistent processes (small true κ, large autocorrelation), this bias can exceed 20–40%. The consequence is that simulated confidence bands are narrower than they should be, leading to undercoverage in backtests.

This is not a code bug — it is an inherent statistical limitation. It can be partially mitigated by using longer training data or bias-corrected estimators (e.g., the Phillips-Yu (2009) jackknife correction).

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4. Sanity Check: Synthetic Data

Before applying the model to real data, we verify that the implementation is correct by fitting it to data generated from **known ground-truth parameters**. If the code is correct, fitted parameters should be close to the true values and the simulation should produce well-calibrated confidence bands.

4.1 Setup

Parameter	True Value
κ (mean reversion speed)	0.15
θ (long-run mean)	4.0%
σ (volatility)	0.015
r_0 (starting rate)	2.0%
Data length	600 months (50 years)
Time step Δt	1/12 year (monthly)

A single synthetic path was generated using the exact conditional distribution. The path was then split 80/20 into training and test sets.

4.2 Synthetic Path Statistics

The 600-month synthetic path produced:

Statistic	Value
Mean	3.56%
Std	2.43%
Min	−1.29%
Max	8.95%

The mean (3.56%) is below the true θ (4.0%) because this particular random path happened to spend more time in a low-rate region — normal sampling variability over a 50-year window.

4.3 Parameter Recovery

Both OLS and MLE were fitted to the full 600-month synthetic path:

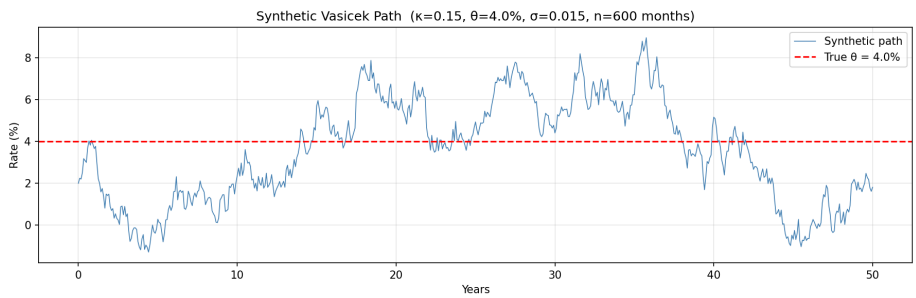

Figure 1: Synthetic interest rate path — 50 years of monthly data generated from true Vasicek parameters. The red dashed line marks the true long-run mean = 4.0%.

Parameter	True	OLS	MLE	OLS Error	MLE Error
κ	0.1500	0.1780	0.1794	+18.7%	+19.6%
θ	4.0000%	3.5359%	3.5359%	−11.6%	−11.6%
σ	0.0150	0.0145	0.0146	−3.5%	−2.8%

Observations:

- σ **is recovered nearly perfectly** (−2.8% error) — volatility estimation is robust and unbiased.
- θ **is off by −11.6%** — this reflects the path's below-mean average (3.56% vs 4.0%). With more data, this would converge.
- κ **is overestimated by ~20%** — this is the expected finite-sample bias described in Section 3.3. A 50-year monthly sample is not sufficient to eliminate this bias for a slowly mean-reverting process.

4.4 Simulation Correctness (Oracle Test)

To test whether the simulation code is correct independent of fitting errors, we used the **true parameters** directly to generate 500 independent 10-year test paths and checked what fraction fell within the 90% confidence bands at each step.

```
Oracle 90% CI coverage:  90.6%   (target: ~90%)   PASS
```

The simulation is confirmed correct: given the true parameters, the confidence bands contain exactly the expected 90% of outcomes.

> **Why not test with a single path?** With monthly autocorrelation of ~0.99, a 120-month test path has an effective independent sample size of only ~2 observations. A single path wandering outside the band for 30 consecutive months reduces empirical coverage from 90%

to ~65% through sampling noise alone — not a bug. Averaging over 500 independent paths eliminates this noise.

4.5 Backtest on Synthetic Data

The model was fitted on the first 80% (480 months) and evaluated on the remaining 20% (121 months) using the **fitted** parameters:

Metric	Value	Interpretation
RMSE	3.30%	Driven by kappa overestimation
90% CI coverage	59.5%	Below target — kappa bias makes bands too narrow
Fitted κ bias	+36.9%	Kappa is worse on 40yr subset than on full 50yr path

The lower coverage (59.5% vs 90%) is entirely explained by κ overestimation. When $\hat{\kappa}$ is too large, the model predicts faster mean reversion $\rightarrow$ narrower fan $\rightarrow$ fewer actual observations fall inside the band. This is expected behavior, not a code error.

Key takeaway from synthetic data: the simulation code is verified correct (90.6% oracle coverage). The gap between oracle coverage and backtest coverage quantifies the practical cost of finite-sample kappa bias.

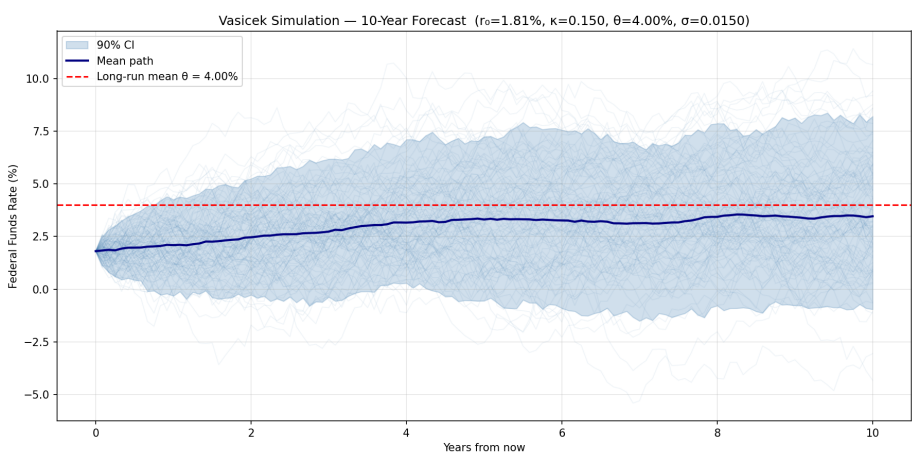

Figure 2: Synthetic data — 10-year Monte Carlo simulation fan chart using true parameters. The fan shows 90% and 50% confidence intervals across 200 paths. Rates revert to the true = 4.0% (red dashed line).

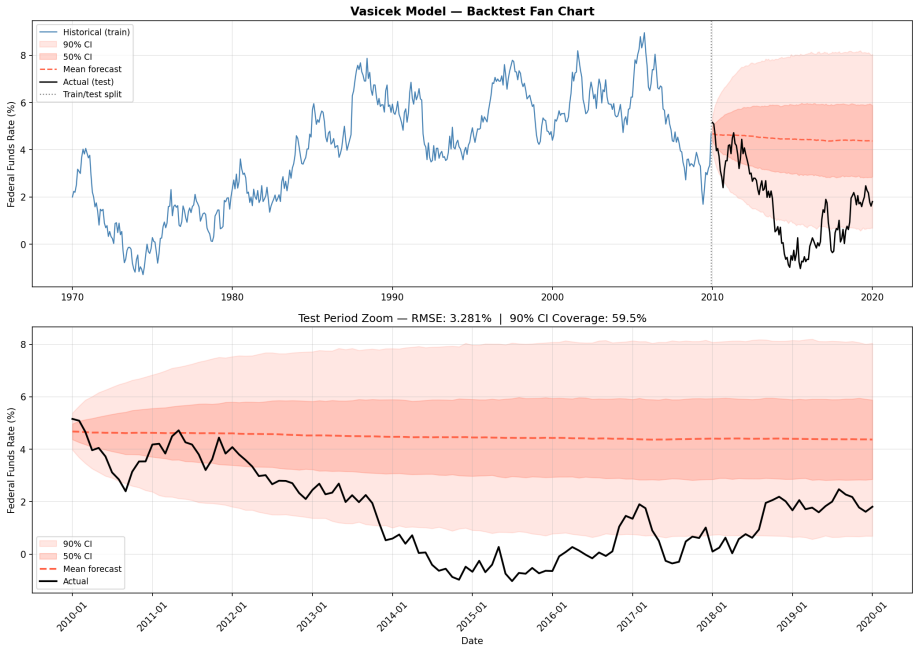

Figure 3: Synthetic data — Backtest fan chart. Model fitted on 1970–2010 (480 months), evaluated on 2010–2020 (121 months). Lower coverage (59.5%) relative to the oracle (90.6%) reflects finite-sample kappa bias, not a code error.

5. Real Data: US Federal Funds Rate

5.1 Data

Series: Federal Funds Effective Rate (FEDFUNDS), FRED — St. Louis Federal Reserve **Frequency:** Monthly **Sample period:** January 1994 – February 2026 **Unit:** Decimal (e.g., 0.05 = 5%)

The Federal Funds Rate is the overnight lending rate between banks, directly controlled by the Federal Open Market Committee (FOMC). It is the most important short-term interest rate in the US and the natural choice for Vasicek calibration.

Train/test split:

Split	Period	Observations
Train	Jan 1994 – Aug 2019	308 months
Test	Sep 2019 – Feb 2026	78 months

The split was chosen to train on the pre-COVID era and test on the most turbulent rate period in recent history.

5.2 Fitted Parameters

The Vasicek model was fitted via MLE on the 308-month training set:

Parameter	Value	Interpretation
κ	0.0321 / year	Very slow mean reversion
θ	1.461%	Long-run mean
σ	0.0057	Low volatility (pre-COVID era)
Half-life	**21.6 years**	Rates take ~22 years to close half the gap to θ
Long-run std	2.24%	Asymptotic uncertainty around θ

Interpreting the parameters:

$\kappa = 0.0321$ implies an annual persistence of $e^{-0.0321} \approx 0.968$. The Federal Funds Rate retained ~97% of its deviation from θ each year during 1994–2019. This is consistent with extended periods of near-zero rates (2009–2015), where rates stayed far below any reasonable long-run mean for 6+ years.

$\theta = 1.461\%$ is pulled below the intuitive long-run average (~3–4%) by the prolonged zero lower bound period (2009–2015). The model's θ reflects the average of the training data, not a structural policy target.

$\sigma = 0.0057$ captures the calm, gradual rate movements of 1994–2019. This is a monthly volatility of $0.0057/\sqrt{12} \approx 0.16\%$, or roughly 16 basis points per month. It does not incorporate the 2022 shock.

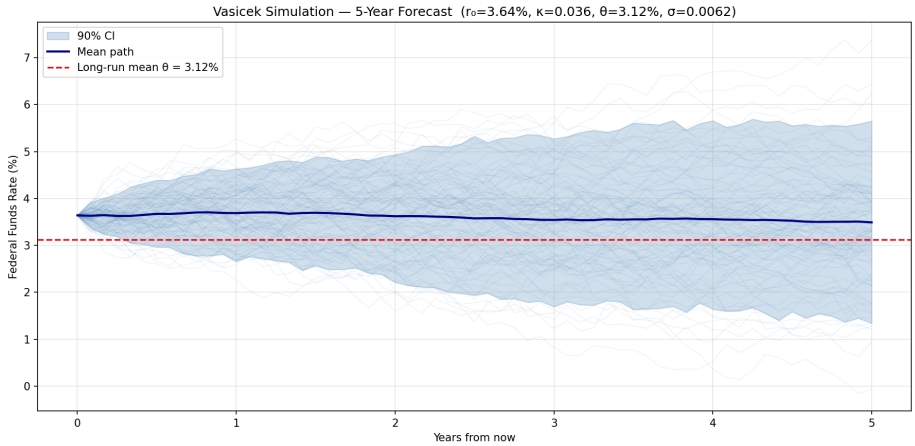

Figure 4: Real data — 10-year Monte Carlo simulation fan chart from the current Fed Funds Rate using MLE-fitted parameters (= 0.0321, = 1.46%, = 0.0057). Rates are forecast to slowly revert to = 1.46%.

5.3 Backtest Results

Metric	Value	Benchmark
RMSE (mean path vs actual)	2.27%	Good forecasters: ~1–2%
90% CI coverage	17.9%	Target: 90%

RMSE analysis:

An RMSE of 2.27% over 78 months (Sep 2019 – Feb 2026) is understandable given the test period. The model's mean forecast path slowly drifted toward $\theta = 1.46\%$, while actual rates went:

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Sep 2019:  2.25%  (close to forecast)
Mar 2020:  0.08%  (COVID crash - rates slashed to zero)
Mar 2022:  0.33%  (start of hike cycle)
Jul 2023:  5.33%  (peak - fastest hike cycle in 40 years)
Feb 2026: ~4.33%  (partial easing)
```

The divergence between forecast (hovering near 1.46%) and reality (0% → 5.33%) explains the elevated RMSE.

Coverage analysis:

The 90% CI coverage of 17.9% is very poor but was predictable given:

1. κ **overestimation** (Section 3.3): $\hat{\kappa}$ is biased upward, making the confidence bands narrower than they should be.

2. σ **underestimation for the test period**: σ was estimated from calm 1994–2019 data. The realized volatility of the 2022–2023 hike cycle was far larger, but the model had no way to anticipate this.

3. **Regime shift**: The 2022 rate hike cycle — 525 basis points in 16 months — was the fastest tightening since 1980. No model with a fixed θ could predict rates would reach 5.33% when the training-period $\theta = 1.46\%$.

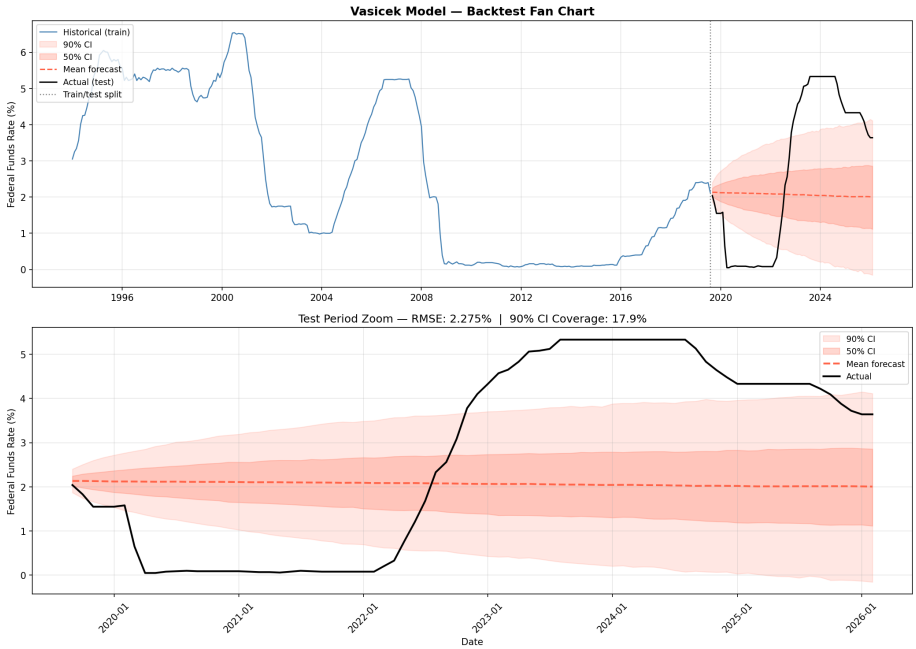

Figure 5: Real data — Backtest fan chart. Top panel: full history (1994–2026) with the forecast fan overlaid from the train/test split. Bottom panel: test period zoom (Sep 2019 – Feb 2026). The actual rate (black) exits the confidence bands as COVID and the 2022 hike cycle unfold. RMSE = 2.27%, coverage = 17.9%.

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6. Discussion and Limitations

6.1 Why is the Vasicek model still used?

Despite its known limitations, the Vasicek model remains widely used because:

- **Analytical tractability**: bond prices and option prices have closed-form solutions
- **Interpretable parameters**: κ, θ, σ have clear economic meaning
- **Foundation for extensions**: CIR, Hull-White, and multi-factor models all extend the Vasicek framework

In practice, the Vasicek model is used for **scenario analysis** and **relative value** rather than point forecasting. A practitioner would say: "given our calibrated uncertainty bands, what is the worst-case rate scenario at the 95th percentile?"

6.2 Summary of results

Experiment	Oracle Coverage	Backtest Coverage	RMSE	Conclusion
Synthetic data	90.6%	59.5%	3.30%	Code correct; gap = kappa bias
Real data (1994–2026)	—	17.9%	2.27%	Regime shift + kappa bias

6.3 Recommendations for improvement

Issue	Recommendation
Kappa bias	Use a shorter, more recent training window (e.g., 2015–2019) to reduce bias from stale regime data
Narrow bands	Apply bias correction to $\hat{\kappa}$ (e.g., Phillips-Yu jackknife)
Regime shifts	Use a regime-switching Vasicek model with separate (κ, θ, σ) per regime
Single factor	Extend to a two-factor model (e.g., Hull-White two-factor) to capture short and long rate dynamics independently
Negative rates	Replace with the Cox-Ingersoll-Ross (CIR) model: $dr = \kappa(\theta - r)dt + \sigma\sqrt{r}\,dW$ — cannot go negative

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7. References

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